



Colour correct

An app that provides you with the best colour palettes possible:
<http://dgit.in/1S5M9Di>



Young guns

Two teenagers managed to hack into a school district's network and take it offline :
<http://dgit.in/1BdpXfX>

DO

in/1AsF3Di. Many 3D printer designs have geared extruders – kindly stay away from those.

3D Printed Parts

Now this is what they call a classic chicken-and-egg situation. The irony of this building-your-own-machine business is that you need 3D-printed parts to build a 3D printer! Most 3D printer vendors also offer 3D printing services, so you can get your parts printed from them.

They take about 20 hours in total to get printed.

Heated bed

Heated beds are needed to print materials like ABS; if the printing surface is not adequately heated, they come out warped. Don't use them when you're starting out though, as maintaining them is a huge hassle.

This is where many users commit mistakes, in that they start using the beds from the word go, as most of the kits come with heated beds. This eventually creates unnecessary complications. So even if you have them, don't use them initially. Instead, print PLA using the masking tape method (Google it!), which, in any case, is more popular and rookie-friendly than ABS.

Slicer software

This software converts your 3D models (in STL format) into instructions on how to use your machine, clearing several doubts such as where to move and how to extrude plastic in .gcode format, among others. Slic3r and cura (<https://software.ultimaker.com/>) are the most popular softwares of the lot.

I recommended the latter as it has got a user-friendly interface and simpler settings. This is the part where you get to learn to play around in the machine settings to get good prints later, so be sure to get well-familiarized with this software. The file formats supported by these softwares are .STL and .OBJ.

Instructions for building the 3D printer

There is no shortage of instructions available on how to assemble a Prusa i3.



Build your 3D printer, get proficient at printing 3D objects and who knows? One day you might get to print a dress for a Victoria's Secret model as well! Isn't that something to strive towards!

You can download the PDF from here:
<http://dgit.in/1cdrRXr>

Here's a guide with interactive animations: <http://dgit.in/1LvqBL4>

You can also follow the 3dprintronics assembly video series: <http://dgit.in/1Es3uvF>

After assembly, you need to program the electronics. The firmware used for this is called Marlin, which needs to be configured.

Here's a guide on how to do that: <http://solidutopia.com/marlin-firmware-user-guide-basic/>. A kit, of course, comes an already configured version.

Notes to remember

- Use tried-and-tested combinations of parts – they're easier to get technical support for. Don't opt for unknown alternatives to save money, especially if you don't fully know what you are dealing with.
- Starting out with a kit is not a bad idea. When I made my own machine, procuring parts was the most difficult fac-

tor and I ended up being forced to import a lot of things like electronics, 3D printed parts and the hot-end.

- The biggest issue you will face when starting out in 3D printing is bed leveling. The bed is the platform on which you print your first whole object. The first layer of sticking is very important. For that, the nozzle in any axis (be it X or Y), should be equidistant from the bed. There are many tutorials which show how to achieve this. Today, there is also the auto-leveling feature which solves this issue, but it comes with its own set of issues.
- Temperatures can be tricky. There is always a standard recommended temperature to print any material – for example, 190 for PLA is the norm. But your setup may vary, based on the accuracy of your thermistor, hot-end design, etc. If the temperature is too high, the prints look like they're half-melted; if it's too low, the layers won't stick and the prints are weaker. So try to keep the temperatures stable.
- Using a fan can improve the print quality dramatically. For example, if pointed towards the printed part, it helps the plastic harden quicker and makes the prints look much better. But be sure to check that your nozzle is not covered with kapton tape or some other material which can cool down because of the fan – not something you would want to happen. You can get a fan duct design from <http://www.thingiverse.com/> which will direct the airflow towards the print.

Getting Help

When in doubt, Google. More often than not, somebody on the Internet has already faced and subsequently solved the problem you're currently experiencing. There is a thriving community on Facebook called '3D Printing – India Working Group' that you can join.

IRC chat for RepRap is also a good place to ask for help: <http://dgit.in/1RfuOXf>.

Happy 3D printing!

Built your own 3D printer? Send us your pictures to editor@dgit.in and get featured here!

Got any queries? Contact us @digitindia and we'll get back to you ASAP. 